

Question

During the hydrolysis of Fe^{3+} , FeOH^{2+} and $\text{Fe}_2(\text{OH})_2^{4+}$ are produced. To investigate the dissociation of $\text{Fe}_2(\text{OH})_2^{4+}$, the following reaction mechanism is proposed:



The first two equations are elementary reactions, and the third is an equilibrium equation. Derive the apparent hydrolysis equilibrium constant $K = \frac{c_1^2}{c_2}$ expressed in terms of k_1 , k_{-1} , k_2 , k_{-2} , c_H , and β . Here, c_H is the hydrogen ion concentration in the aqueous solution, c_1 is the total concentration of mononuclear ions, and c_2 is the concentration of the dimer $\text{Fe}_2(\text{OH})_2^{4+}$. The concentration of water is already included in the constants and need not be explicitly stated.

A.
$$K = \frac{k_1 + k_2 c_H}{\frac{c_H}{c_H + \beta} \left(k_{-1} \frac{c_H}{c_H + \beta} + k_{-2} \frac{\beta}{c_H + \beta} \right)}$$

B.
$$K = \frac{k_1 + k_2 c_H}{k_{-1} \frac{c_H}{c_H + \beta} + k_{-2} \frac{\beta}{c_H + \beta}}$$

C.
$$K = \frac{k_1 + k_2 c_H}{\frac{c_H}{c_H + \beta} (k_{-1} + k_{-2})}$$

D.
$$K = \frac{(k_1 + k_2 c_H)(c_H + \beta)^2}{k_{-1} c_H^2}$$

E.
$$K = \frac{(k_1 + k_2 c_H)(c_H + \beta)^2}{k_{-2} c_H \beta}$$

F.
$$K = \frac{k_1 + k_2 c_H}{\frac{\beta}{c_H + \beta} \left(k_{-1} \frac{c_H}{c_H + \beta} + k_{-2} \frac{\beta}{c_H + \beta} \right)}$$

G.
$$K = \frac{k_1 + k_2 c_H}{\frac{\beta}{c_H + \beta} \left(k_{-1} \frac{\beta}{c_H + \beta} + k_{-2} \frac{c_H}{c_H + \beta} \right)}$$

Answer

Correct Answer: **G**

Detailed Explanation

The rate of the first depolymerization reaction is $v_{d1} = k_1 c_2$

CHECKPOINT

1 PTS

$$v_{d1} = k_1 c_2$$

The rate of the second depolymerization reaction is $v_{d2} = k_2 c_2 c_H$

CHECKPOINT

1 PTS

$$v_{d2} = k_2 c_2 c_H$$

The total depolymerization rate is $v_d = v_{d1} + v_{d2} = (k_1 + k_2 c_H) c_2$

CHECKPOINT

0.5 PTS

$$v_d = (k_1 + k_2 c_H) c_2$$

The equilibrium concentrations of mononuclear species are derived using the distribution coefficient:

$$c(\text{Fe}^{3+}) = c_1 \frac{c_H}{c_H + \beta} c(\text{FeOH}^{2+}) = c_1 \frac{\beta}{c_H + \beta}$$

CHECKPOINT

0.5 PTS

$$c(\text{Fe}^{3+}) = c_1 \frac{c_H}{c_H + \beta}$$

CHECKPOINT

0.5 PTS

$$c(\text{FeOH}^{2+}) = c_1 \frac{\beta}{c_H + \beta}$$

The rate of the first polymerization reaction is $v_{a1} = k_{-1}c(\text{FeOH}^{2+})^2 = k_{-1}c_1^2\left(\frac{\beta}{c_H + \beta}\right)^2$

CHECKPOINT

1 PTS

$$v_{a1} = k_{-1}c_1^2\left(\frac{\beta}{c_H + \beta}\right)^2$$

The rate of the second polymerization reaction is $v_{a2} = k_{-2}c(\text{Fe}^{3+})c(\text{FeOH}^{2+}) = k_{-2}c_1^2\frac{c_H}{c_H + \beta}\frac{\beta}{c_H + \beta}$

CHECKPOINT

1 PTS

$$v_{a2} = k_{-2}c_1^2\frac{c_H}{c_H + \beta}\frac{\beta}{c_H + \beta}$$

The total polymerization rate is $v_a = v_{a1} + v_{a2} = \left(k_{-1}\left(\frac{\beta}{c_H + \beta}\right)^2 + k_{-2}\frac{c_H}{c_H + \beta}\frac{\beta}{c_H + \beta}\right)c_1^2$

CHECKPOINT

0.5 PTS

$$v_a = \left(k_{-1}\left(\frac{\beta}{c_H + \beta}\right)^2 + k_{-2}\frac{c_H}{c_H + \beta}\frac{\beta}{c_H + \beta}\right)c_1^2$$

At equilibrium, the total depolymerization rate v_d of the dimer equals the total polymerization rate v_a , i.e.,

CHECKPOINT

1 PTS

$$v_d = v_a$$

$$(k_1 + k_2c_H)c_2 = \left(k_{-1}\left(\frac{\beta}{c_H + \beta}\right)^2 + k_{-2}\frac{c_H}{c_H + \beta}\frac{\beta}{c_H + \beta}\right)c_1^2$$

Rearranging gives

$$K = \frac{k_1 + k_2 c_H}{\frac{\beta}{c_H + \beta} \left(k_{-1} \frac{\beta}{c_H + \beta} + k_{-2} \frac{c_H}{c_H + \beta} \right)}$$

CHECKPOINT

0.5 PTS

$$K = \frac{k_1 + k_2 c_H}{\frac{\beta}{c_H + \beta} \left(k_{-1} \frac{\beta}{c_H + \beta} + k_{-2} \frac{c_H}{c_H + \beta} \right)}$$